

USING AI IN QUANT FINANCE

SESSION 3

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TODAY

Who benefits

Working with AI on Python

Failures that raise no error

From scripts to systems

Teaching

WHO BENEFITS

YVES HILPISCH ON CODING

Programming is still the engine of quantitative finance, and GenAI only amplifies – not replaces – the need for precise, testable code …

hilpisch.com, summarizing Why Coding Skills Still Matter in the Age of AI, by The Python Quants with GPT-5, dated May 29, 2025

THE SAME ASSISTANT, TWO USERS

Expert

- Asks for specific functions
- Recognizes when output needs changes
- Spots outdated formulas and assumptions
- Suspects a strong backtest

Novice

- Asks for a whole strategy
- Runs code without being able to judge it
- Asks the AI to fix errors it cannot diagnose
- Accepts a strong backtest

Hilpisch (2025). His expert and novice cases are illustrative scenarios.

FOUR KINDS OF PRACTITIONER

Type	Skills	Hilpisch's expectation
Super quants	Finance, Python, and AI tools	One may do the prototyping work of several junior analysts
Traditional quants	Finance and some coding, no AI tools	Effective, but slower
Non-coding domain experts	Finance only	Limited to tools others prepare
Non-domain tech experts	Technology only	Need finance partners to avoid misapplying AI

Hilpisch (2025), section 6

REPORTS FROM AFTER DECEMBER 2025

Source	Report
Tal Fishman, Lord Abbett	“The error rate from a vague prompt used to be 70–80%. In December that flipped.”
Ken Griffin, Citadel	Work done by finance PhDs over weeks or months is now done by agents in hours or days
Alejandro Lopez-Lira, Florida	AI systems are “at the level of a fourth-year Ph.D. student”

AI Street: Fishman, March 3, 2026; Griffin, May 21, 2026; Lopez-Lira, July 28, 2026. Hilpisch’s paper predates this period.

A RANDOMIZED TRIAL ON LEARNING

50%
quiz score with AI assistance

67%
quiz score coding by hand

2 min
faster with AI, not statistically significant

- 52 mostly junior engineers learning a new Python library. Debugging showed the largest gap.
- Those who asked for explanations or conceptual help scored higher than those who delegated.
- The assistant was a sidebar, not a coding agent. The quiz came right after the task.

Shen and Tamkin (2026), How AI assistance impacts the formation of coding skills, Anthropic.

WORKING WITH AI ON PYTHON

WHAT THE AGENT DOES

1. Writes code and saves it to a file.
2. Runs it with a terminal command.
3. Reads the output.
4. Revises and repeats.

Common uses in a quant project: boilerplate, explaining code or math, suggesting vectorization, and finding how a library works.

Uses from Hilpisch (2025)

ARITHMETIC AND CODE

- Levy changed the last digit of accounting figures, for example $7.334 billion to $7.335 billion.
- GPT-4's accuracy in predicting earnings changes fell from 60% to chance.
- Having the model write and run code for the arithmetic improved accuracy.

> Prompt: "Compute this in Python."

Bradford Levy, University of Chicago, "Caution Ahead," reported in AI Street, January 12, 2025. The model is an older one.

BEFORE IT STARTS

```
Plan how to do this before you start.
```

```
Tell me what you are about to do and what you are assuming.
```

- Planning brings relevant knowledge into the conversation and creates steps you can check.
- Given an ambiguous request, it answers confidently instead of asking.

HOW THE MODEL BEHAVES

Knowledge in use

If it said last week that method A beats method B, but that is not in this conversation, it will probably use whichever appeared more in its training data.

Put the method you want in the conversation.

Momentum

After a roadblock it finds a way around, then another. It rarely asks whether a different path was better.

Restate the goal, not the obstacle.

A PROJECT INSTRUCTIONS FILE

```
# CLAUDE.md

Fundamentals: as-reported values, dated by filing date.
Universe: include delisted companies and delisting returns.
Time series: never shuffle. Train on the past, test on the future.
Returns: state costs and borrow assumptions in every backtest.
Numbers: compute in code. Set and log random seeds.
Runs: after each script, add the script, date, and output to RUNS.md.
```

Read at the start of every session in the folder. When you settle something, tell the AI to add it.

RECORDS FOR A RESEARCH PROJECT

File	Holds	Why
`CLAUDE.md`	What is always true	Read at the start of every session
`RUNS.md`	Script, date, and output of each run	Kept apart so it is not read every time
`memory/2026-11-05.md`	Decisions, rejected options, and the reasons	Each new session starts knowing nothing
Git	Every version of the code	The AI can run it and write the commit messages
Parquet files	Data	Fast, small, and left out of Git

FAILURES THAT RAISE NO ERROR

CODE THAT RUNS AND IS WRONG

Failure	What happens
Restated fundamentals	A 2009 portfolio uses numbers restated in 2011
Fiscal period end instead of filing date	Weeks or months of free information
Only surviving companies	Every past portfolio holds firms known to survive
No delisting returns	A company that goes to zero is not recorded at zero
Shuffled cross-validation	The model trains on 2018 to predict 2006
No costs or borrow	The short side is free

A QUESTION THE SCHEMA CANNOT ANSWER

> Count distinct tickers with data in 2005. Then count how many of those still have data in the most recent year, and show me some that do not.

If the two counts match, the data contains only survivors.

INDEPENDENT CHECKS

AI-written code	Check against
Monte Carlo option pricer	Closed form, and convergence as paths increase
Greeks	Finite differences
Portfolio optimizer	Two-asset closed form
Data pipeline	A few rows by hand
Positive backtest result	Permutation test

First two from Hilpisch (2025)

EXERCISE: PRICE A CALL

> Write Python that prices a European call by Monte Carlo under geometric Brownian motion: spot 100, strike 110, one year, rate 5%, volatility 20%. Report the standard error. Compare the price with Black-Scholes. Estimate delta by finite differences and compare it with the closed form.

6.04

Black-Scholes price

0.45

Black-Scholes delta

Parameters from Hilpisch (2025)

TESTING MORE IDEAS

- Fishman: projects that took weeks can be attempted in days or hours, so the firm attempts more of them.
- WorldQuant's student competition: more than 150,000 individual registrants in 2026, about double 2025.
- Exercise: have a model generate a few hundred candidate alphas, evaluate them, then apply a multiple-testing correction.

AI Street: Fishman, March 3, 2026; WorldQuant, August 27, 2026

AN AUDIT OF 19 LLM TRADING STUDIES

2

report an extractable time-consistent split

1

reports an explicit transaction-cost model

1

documents universe or survivorship handling

15 of the 19 are at the lowest reproducibility tier. None reach the highest.

Yao and Zheng (2026), "Beyond Agent Architecture: Execution Assumptions and Reproducibility in LLM-Based Trading Systems," arXiv 2606.08285.

FROM SCRIPTS TO SYSTEMS

KEEP WHAT WORKED

> `Create a skill for this, and save the code as a script that the skill uses.`

- The next run uses tested code. The model does not write it again.
- Pisaneschi, CFA Institute: when the model makes a mistake, add the “gotcha” to the skill file.
- Test the skill on a case you can verify before relying on it elsewhere.

Brian Pisaneschi in AI Street, April 7, 2026

SPLIT THE WORK, THEN CHECK IT

- Lopez-Lira's portfolios separate macro analysis, company research, portfolio construction, and verification.
- Splitting keeps each context small. A full context degrades performance.
- A new agent checks the work: within the same session, "the model is biased by what has already been written."
- Agents are defined in Markdown files. An instruction added once applies to all future runs.

Alejandro Lopez-Lira in AI Street, July 28, 2026

WHERE THE LLM SITS

1. The model turns text into a number. Conventional code decides what it means.
2. The expensive model step is shared. Firms differ on the cheap steps after it.
3. Signals are formed without access to outcomes, enforced by what the code imports.
4. A positive R-squared is a hypothesis until it survives a permutation test.

> High-risk finance processes suit workflows with checks at each step, not open-ended agents (Pisaneschi).

TEACHING

HILPISCH'S RECOMMENDATION

A student should graduate knowing how to code and how to use AI to help code, knowing the strengths and pitfalls of each.

Hilpisch (2025). His related story, The Maestro and the Apprentice, asks how an unprepared manager directs powerful tools.

QUESTIONS FOR QUANT PROGRAMS

1. Is Python a prerequisite, or something students learn while using AI?
2. Which code must students write without AI, and when?
3. Should exams test reading and debugging code more than writing it?
4. How do students reach the expert column if AI does the novice work?
5. What should a quant thesis disclose about AI use?



QUESTIONS?